

# Sathyaseelan K

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## Summary

Final-year IT undergraduate with hands-on experience building scalable backend systems, REST APIs, and full-stack applications across Java, Python, and the GCP/Supabase stack. Seeking Software Engineering / Backend roles building scalable, data-driven solutions.

## Education

### Bachelor of Technology in Information Technology

M. Kumarasamy College of Engineering, Karur, TN  
Aug 2023 – Present • CGPA: 7.9/10.0

### Higher Secondary Certificate (HSC)

Chelammal Vidhyaashram, Trichy, TN  
Jun 2022 – Jun 2023 • Score: 86%

## Technical Skills

|                       |                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------|
| Programming Languages | Java (Proficient), Python (Intermediate), SQL (Proficient), C (Basic)               |
| Web Technologies      | HTML, CSS, Flask                                                                    |
| Databases & Backend   | MySQL, Supabase, MongoDB                                                            |
| Core Subjects         | Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks       |
| AI / ML               | Machine Learning, Computer Vision, OpenCV, TensorFlow, NLP                          |
| Tools & Platforms     | Git, GitHub, Postman, IntelliJ IDEA, VS Code, Android Studio, Google Cloud Platform |

## Experience

### Google Gemini Student Ambassador

May 2026 – Present

Google Campus Ambassador Program

- Spearheaded campus-wide adoption of Google's Gemini AI by organizing technical workshops and coding sessions, empowering students to build AI-integrated applications.
- Mentored peers on prompt engineering, API integrations, and leveraging Large Language Models (LLMs) to optimize modern software development workflows.

## Projects

### Megam – AI-Powered Global Air Quality Monitoring Platform (Live | Code)

Nov – Dec 2025

React, TypeScript, Flask, Python, scikit-learn, NASA MODIS

- Engineered a Python/Flask data pipeline integrating IQAir and NASA MODIS APIs to successfully aggregate and process live metrics across 80,000+ monitoring stations.
- Developed scikit-learn predictive models and a React/TypeScript front-end with a Three.js 3D globe, enabling users to interactively visualize and forecast pollution trends globally.

### Re:Vive – AI, NLP and AR Cultural Discovery Platform (Live | Code)

Jun – Dec 2025

Flask, NLP, Supabase, JavaScript, AR Integration

- Architected a Flask-based backend utilizing NLP techniques to generate personalized landmark recommendations and multilingual narration, significantly enhancing cultural accessibility.
- Integrated AR-based discovery features backed by Supabase; the system's architecture and findings were peer-reviewed and published at the IEEE ICCCES 2026 conference.

### IoT-Based Active Peripheral Security and Automated Attendance (Code)

Feb – Mar 2026

ESP32, RC522 RFID, INA219, Flutter, MySQL

- Engineered an ESP32-based hardware framework using RC522 RFID and a 5V relay circuit to instantly block unauthorized peripheral access, successfully securing physical endpoints.
- Built a cross-platform Flutter app and MySQL database for live monitoring, completely replacing manual attendance registers with 100% automated, tamper-proof digital records.

### LUDO – AI-Based Personal Decision Assistant (Code)

Dec 2025 – Jan 2026

Python, Rule-Based Logic, Contextual APIs

- Developed a Python AI application using rule-based logic and contextual APIs to accurately interpret and execute natural language commands with minimal processing latency.
- Architected the system using highly modular components, creating a scalable foundation that allows for rapid integration of future mobile interfaces and automation workflows.

## Research Publication

- IEEE Publication:** "Re:Vive: An Intelligent Interactive Platform for Cultural Discovery through AI, NLP, and AR," ICCCES 2026.

## Certifications

- Oracle Certified:** Data Science Professional; Cloud Database Services (Oct 2025).
- Coursera:** Supervised Machine Learning – DeepLearning.AI & Stanford Online (Sep 2025).
- Coursera:** Cybersecurity for Everyone – University of Maryland (Feb 2025).